

The term $\rho V^2/2$ in the Bernoulli equation is named as _____

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Choices:

☐ hydrostatic pressure

☐ total pressure

☐ static pressure

☐ dynamic pressure

correct

Which mathematician formulated and applied the buoyancy principle in history's first nondestructive test to determine the gold content of the crown of a king?

page 4

Choices:

☐ Galileo

☐ Pascal

☐ Aristotle

☒ Archimedes

correct

Question: 4 (Marks: 1)

The flow in which all fluid variables remain constant in time is called _____

Choices:

☐ uniform flow

☐ non-uniform flow

☒ steady flow

correct

☐ unsteady flow

Question: 3 (Marks: 1)

By the Pascal's law the pressure applied to a confined fluid increases the pressure throughout by _____

pg#27

Choices:

☐ the larger amount

☒ the same amount

correct

☐ the shifting amount

☐ the smaller amount

Question 2 (Mark: 1)

The solid body dropped in the fluid will _____ if the density of a solid body is smaller than the density of the fluid.

Choices:

☐ oscillate

☐ suspend

☐ sink

☒ float

correct

Question: 6 (Marks: 1)

In fluids, stress is proportional to _____

pg#1

Choices:

strain



strain rate



correct

displacement



force



Question: 7 (Marks: 1)

The gas flows are assumed to be incompressible for Mach number _____

pg#75

Choices:

☒ < 0.3

correct

☐ > 0.3

☐ < 0.5

☐ > 0.5

Question: 8 (Marks: 1)

The fully developed flow in a circular pipe, in cylindrical coordinate system, is ____

pg#11

Choices:

☒ one dimensional

correct

☐ two dimensional

☐ three dimensional

☐ zero dimensional

Question: 9 (Marks: 1)

Any physical quantity can be characterized by _____

pg#14

Choices:

dimensions

correct

units

SI system

English units

Question: 11 (Marks: 1)

The correct relation for mass, density and volume is _____

$$(V=M/d) \text{ or } (d=M/V)$$

Choices:

☐ density = mass + volume

☐ density = mass - volume

☒ density = mass / volume

correct

☐ density = mass * volume

Question: 10 (Marks: 1)

A liquid is said to wet the surface when contact angle _____

pg#24

Choices:

☐ $\phi < 90^\circ$

correct

☐ $\phi = 90^\circ$

☐ $\phi > 90^\circ$

for not wet

☐ $\phi = 0^\circ$

Question: 10 (Marks: 1)

A liquid is said to wet the surface when contact angle _____

Choices:

☒ $\phi < 90^\circ$

correct

☐ $\phi = 90^\circ$

☐ $\phi > 90^\circ$

☐ $\phi = 0^\circ$

Question: 9 (Marks: 1)

Any physical quantity can be characterized by _____

Choices:

dimensions



correct

units



SI system



English units



Question: 13 (Marks: 1)

Flow visualization is useful not only in physical experiments but also in _____ as well.

pg#34

Choices:

☒ numerical solutions

correct

☐ analytical solutions

Question: 14 (Marks: 1)

_____ acceleration is zero for steady flows.

local acceleration
is non zero for
unsteady flow

Choices:

☐ local

☐ advective

☒ convective

☐ normal

correct

Question: 14 (Marks: 1)

_____ acceleration is zero for steady flows

Choices:

local



advective



convective



correct

normal



Question: 17 (Marks: 1)

The Reynolds transport theorem provides the link between the system and _____ approaches.

Choices:

☒ control volume

correct

☐ control mass

☐ control surface

☐ control area

Question: 16 (Marks: 1)

$\vec{\omega} = \frac{\psi}{2} \vec{\zeta}$, $\vec{\omega}$ is rotation vector and $\vec{\zeta}$ is vorticity vector.

psi/2

Choices:

☐ $\vec{\omega} = \frac{\psi}{2} \vec{\zeta}$

correct

☐ $\vec{\omega} = \frac{\psi}{3} \vec{\zeta}$

☐ $\vec{\omega} = 2\vec{\zeta}$

☐ $\vec{\omega} = 3\vec{\zeta}$

Question: 15 (Marks: 1)

Which one of the following represent local part of pressure (P)?

Choices:

☐ $(\vec{V} \cdot \vec{V}) P$

☐ $(\vec{V} \cdot \vec{V}) P$

☐ $\frac{\partial P}{\partial t}$

☐ ∇P

Question: 15 (Marks: 1)

Which one of the following represent local part of pressure (P)?

Choices:

☐ $(\vec{V} \cdot \vec{V}) P$

☐ $(\vec{F} \cdot \vec{V}) P$

☐ $\frac{dP}{dt}$

☐ ∇P

Question: 22 (Marks: 1)

The Bernoulli equation is valid in regions of _____ flow.

Choices:

☒ steady, incompressible

correct

☐ unsteady, incompressible

☐ steady, compressible

☐ unsteady, compressible

For some extensive property B or its corresponding intensive property b , the material derivative is given as _____

Choices:



Question: 29 (Marks: 3)

What is stagnation pressure? Write it mathematically

Answer:

$P_0 = P + \frac{\rho V^2}{2}$

Question: 30 (Marks: 5)

Determine the density, and mass of gasoline in a rectangle-shaped container whose dimensions are $2\text{ m} \times 3\text{ m} \times 4\text{ m}$ at 100 kPa and 0°C and specific gravity is 0.7 .

Answer:

Question: 31 (Marks: 3)

Consider the following steady, two dimensional velocity field given as $V = (u, v) = y^2 i + 3x^2 j$. Find an analytical expression for the flow streamlines.

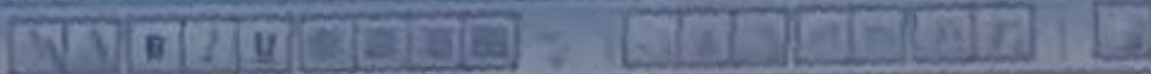
Answer:

$\frac{1}{2}y^3 + x^3 = C$

Question: 28 (Marks: 3)

Give an example illustrating the importance of choosing the most appropriate coordinate system for dimensionality.

Answer:



Question: 27 (Marks: 1)

Which of the following represents the x-component of the rate of rotation vector?

Choices:

Regard to Ali G
Remember in your precious prayers